SDG BLOCKCHAIN ACCELERATOR

Mentor Feedback Report ; SALA

Program: EMURGO ; UNDP Blockchain Accelerator (Cohort 2)
Project: SALA ; Digital Identity & Credential Verification
Partners: UNDP Bosnia and Herzegovina, University of Sarajevo
Mentorship Focus: Identity Architecture, Credential Integrity, Institutional Adoption
Total Mentoring Sessions: 6
Mentor(s): Ahmed Hadded/ Viet Hung

1. Initial Context and Entry Assessment

SALA entered Cohort 2 with a problem that is both real and well understood: academic credential fraud, manual verification processes, and the lack of a trusted, digital-native way to verify educational records across institutions and borders.

From a problem-definition perspective, SALA was one of the strongest teams in the cohort. The need was clear, the users were identifiable (universities, students, employers), and the institutional context, particularly in Bosnia and Herzegovina, was well articulated.

However, from an identity architecture perspective, the project entered the accelerator with a noticeable tension:

- On one hand, the team clearly understood the importance of privacy, selective disclosure, and institutional trust.
- On the other hand, the initial technical direction leaned toward an NFT-based credential model, which, while practical for a PoC, does not align cleanly with established identity and credential standards.

Mentoring therefore focused less on *whether* SALA should exist, and more on *how far the initial implementation could responsibly go without locking the project into a suboptimal long-term architecture*.

2. Core Architectural Tension: NFTs vs Verifiable Credentials

One of the most important, and recurring, topics throughout mentoring was the choice to begin with an NFT-based credential model rather than a W3C Verifiable Credentials (VC) approach.

Why This Was a Problem (Mentor Perspective)

From an identity systems standpoint, academic credentials are:

- Assertions, not assets
- Issued by institutions, not owned in a transferable sense

- Meant to be selectively disclosed, not publicly enumerated

NFTs, by contrast:

- Are inherently asset-like
- Imply ownership semantics that do not map cleanly to education
- Encourage public visibility unless carefully constrained

This mismatch was discussed early and often. The mentor position was not that NFTs are "wrong," but that they are a poor default for identity unless carefully scoped.

That said, the mentoring also recognized two practical realities:

1. The team needed a working PoC within accelerator timelines
2. Institutional stakeholders (universities) often understand "certificate issuance" more easily than abstract identity standards

The mentoring goal therefore became:
Use NFTs as an anchoring mechanism, not as the identity model itself.

3. Mentoring Sessions and Evolution

Session 1 ; Problem Framing and Institutional Reality

The first session focused almost entirely on **context**, not code.

We discussed:

- How universities currently issue and verify credentials
- Who actually bears the cost of verification today
- Where fraud and inefficiency really occur
- What universities are *willing* to change versus what they are not

A key insight from this session was that universities care more about control and legitimacy than technology. Any solution perceived as undermining institutional authority, even unintentionally, would face resistance.

This session also surfaced an important strength: SALA had already begun building a real relationship with the University of Sarajevo, rather than treating universities as hypothetical users.

Mentor feedback emphasized that this relationship was more valuable than any technical prototype at this stage.

Session 2 ; First Architecture Review (NFT-Based PoC)

The second session reviewed the initial NFT-based credential design.

Technically, the design was sound:

- Credential data hashed
- Hash anchored on Cardano
- Documents stored off-chain
- QR-based verification flow

However, the mentor feedback was deliberately critical:

- NFTs were doing too much conceptual work
- Ownership semantics were unclear
- Revocation and updates were not naturally expressed
- Selective disclosure was effectively impossible

The mentor position was clear:

> This works as a proof of anchoring, not as a proof of identity.

The team accepted this critique and agreed to explicitly frame the NFT as a verification anchor, not as the credential itself.

Session 3 ; Privacy, Disclosure, and “What Should Never Be On-Chain”

This session focused on privacy and long-term risk.

We examined:

- What data could safely be anchored on-chain
- What data should never be on-chain
- How QR verification might leak information unintentionally
- How to avoid creating a global index of student credentials

We pushed hard on privacy-by-design, particularly given EU data protection expectations and university sensitivities.

Key outcomes:

- Clear separation between document storage and on-chain anchors
- Agreement that no personal data belongs on-chain

- Recognition that even metadata can become sensitive over time

This session marked a turning point where the team began thinking less like “blockchain builders” and more like identity system designers.

Session 4 ; Verifiable Credentials and Future Architecture

Only after stabilizing the PoC did mentoring move decisively toward Verifiable Credentials.

This session focused on:

- W3C VC concepts (issuer, holder, verifier)
- How SALA’s current PoC maps, or fails to map, to VC flows
- The role of DIDs and trust registries
- Cardano-native identity options (including Veridian)

The mentor stance was intentionally firm:

> An NFT-based system cannot be the end state for academic credentials.

At the same time, the mentor acknowledged:

- That jumping directly to full VC infrastructure would have slowed progress
- That institutional partners were not yet ready to reason about VCs

The agreed position was pragmatic:

- Keep the current PoC as an anchoring and demonstration tool
- Explicitly design the next iteration around VCs
- Treat NFTs as an implementation detail, not a protocol commitment

Session 5 ; Institutional Adoption and the University of Sarajevo

This session shifted almost entirely away from technology.

We reviewed:

- How SALA engaged with the University of Sarajevo
- How requirements were gathered
- What language resonated with institutional stakeholders
- What caused hesitation or concern

Mentor feedback here was largely positive.

The team demonstrated:

- Respect for institutional processes
- Willingness to adapt terminology and framing
- Patience in building trust rather than pushing technology

This was arguably SALA's strongest area. While the technical architecture was still evolving, the institutional relationship was real, active, and productive.

The mentor explicitly noted that many technically stronger teams fail at this step.

Session 6 ; Readiness, Limitations, and Honest Positioning

The final session focused on realism.

We discussed:

- What the current PoC does well
- What it does not solve
- How it should be presented to UNDP and universities
- Risks of overselling the solution

Feedback emphasized the importance of honest positioning:

- This is a verification PoC, not a full identity system
- NFTs are scaffolding, not the destination
- VCs are the architectural future, but not yet implemented

The team agreed to document limitations explicitly rather than hiding them.

4. Overall Mentor Assessment

SALA made meaningful progress during Cohort 2, but not in a linear "feature delivery" sense.

The biggest achievement was not the NFT PoC itself, but:

- The clarification of architectural direction
- The avoidance of early lock-in to a flawed identity model
- The establishment of a credible institutional partnership

SALA showed maturity in being willing to hear uncomfortable feedback, particularly around NFTs and identity, and adjusting course accordingly.

5. Strengths Observed

- Strong problem understanding
- Serious engagement with institutional partners
- Willingness to rethink early technical assumptions
- Good instinct for privacy and data protection

6. Weaknesses and Risks

- Initial over-reliance on NFT metaphors for identity
- Delayed engagement with VC standards
- Risk of confusion if PoC is misinterpreted as final architecture
- Need for deeper identity expertise in future phases

7. Mentor Conclusion

SALA should not be judged primarily as a technical identity solution yet. It should be judged as a relationship-building and architectural learning exercise that successfully avoided early mistakes many identity projects make.

The decision to start with an NFT-based PoC was not ideal, but it was mitigated by:

- Clear recognition of its limitations
- Willingness to pivot toward Verifiable Credentials
- Strong grounding in institutional reality

If SALA continues into a next phase, mentoring should shift decisively toward:

- W3C Verifiable Credentials
- DID governance models
- Trust registries and issuer accreditation
- Long-term interoperability across institutions

Handled carefully, SALA has the potential to evolve into a serious credential verification system. Handled carelessly, it risks becoming yet another NFT-based credential experiment with limited longevity. The difference will come from discipline in the next phase.